DRAWING DESCRIPTIONS — PATENT C

Method and System for Autonomous Self-Regulation and Cognitive Resynchronization in Neural Processing Systems During Disconnection from External Control Layers

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a system architecture block diagram showing the neural processing system with the cognitive control layer, fallback controller, and the transition mechanism between connected and autonomous operation.

FIG. 2 is a state transition diagram showing the three operational states and the conditions that trigger transitions between them.

FIG. 3 is a graph showing the energy-aware modulation curve that governs autonomous self-regulation behavior across four energy regions.

FIG. 4 is a signal diagram showing the output waveform of the autonomous

input generator including circadian base signal and attention bursts.

FIG. 5 is a structural diagram of the resynchronization payload transmitted to the cognitive control layer upon reconnection.

FIG. 6 is a set of timeline diagrams showing three recovery scenarios: normal reconnection, crash recovery, and clean shutdown.

FIG. 7 is an end-to-end signal flow diagram showing the complete system during both connected and autonomous operation, illustrating how the fallback mechanism seamlessly replaces external cognitive modulation with internal energy-aware self-modulation without interrupting neural processing or energy harvesting.

DETAILED DESCRIPTION OF THE DRAWINGS

Reference Numeral Convention: Each figure in this patent uses a per-figure numbering series (FIG. 1 uses 100-series, FIG. 2 uses 200-series, through

FIG. 7 uses 700-series), with even-numbered increments. Where the same physical component appears in multiple figures, it receives the numeral appropriate to that figure's series.

FIG. 1 — System Architecture with Fallback

FIG. 1 is a system architecture block diagram showing the complete system including the fallback mechanism. The diagram is arranged in three vertical sections.

The top section depicts an external cognitive control layer 100 drawn as a cloud shape and labeled "External Cognitive Control Layer (e.g., Claude)." The external cognitive control layer 100 provides intelligent cognitive decisions to the system. Two downward arrows exit the external cognitive control layer 100: a first arrow carrying modulation commands in the range -1.0 to +1.0, and a second arrow carrying a heartbeat signal implicit via tool calls. A dashed outline around the external cognitive control layer 100

indicates that it may become unavailable.

The middle section contains a heartbeat watchdog 102 depicted as a rectangle.

The heartbeat watchdog 102 monitors the time elapsed since the last cognitive input, with a timeout threshold of 30 seconds and a check interval of 5 seconds. Two output paths exit the heartbeat watchdog 102: a solid left arrow labeled "Heartbeat active" leading to connected mode, and a dashed right arrow labeled "Heartbeat timeout" leading to autonomous mode.

On the left path, a cognitive-driven operation block 104 is depicted as a rectangle. The cognitive-driven operation block 104 receives input controlled by Claude in the range 0 to 1, and modulation controlled by Claude in the range -1 to +1. An arrow from the cognitive-driven operation block 104 proceeds downward to the neural processing section.

On the right path, an autonomous fallback controller 106 is depicted as a rectangle. The autonomous fallback controller 106 generates self-produced

input comprising a circadian base signal plus stochastic attention bursts, and applies energy-aware modulation in the range -0.4 to +0.3. An arrow from the autonomous fallback controller 106 proceeds downward to the neural processing section. Additionally, the autonomous fallback controller 106 outputs to a step buffer 108 depicted as a rectangle and to a state snapshot store 110 depicted as a cylinder or disk shape.

The bottom section contains a spiking neural network 112 and an energy harvester 114, depicted together as a large rectangle labeled "Spiking Neural Network + Energy Harvester." The spiking neural network 112 and energy harvester 114 provide continuous operation regardless of whether the control source is the cognitive-driven operation block 104 or the autonomous fallback controller 106. A key annotation states: "Seamless transition: SNN continues processing without interruption. The control source changes; the computation continues."

FIG. 2 — State Transition Diagram

FIG. 2 is a finite state machine diagram with three states and labeled transitions between them.

A connected state 200 is depicted on the left with a bold outline. The connected state 200 is labeled "CONNECTED" with sublabel "Cognitive layer active." Inside the connected state 200: "Input from cognitive layer" and "Modulation from cognitive layer." A start indicator points to the connected state 200 identifying it as the initial state.

An autonomous state 202 is depicted on the right. The autonomous state 202 is labeled "AUTONOMOUS" with sublabel "Fallback active." Inside the autonomous state 202: "Self-generated input," "Energy-aware modulation," and "State buffering active." An annotation notes that snapshots are taken every 50 steps.

A recovering state 204 is depicted at the bottom. The recovering state 204 is labeled "RECOVERING" with sublabel "Resynchronization." Inside the recovering state 204: "Transmitting buffer" and "Restoring cognitive control."

A first transition arrow 206 from the connected state 200 to the autonomous state 202 is labeled "Heartbeat timeout (>30s no tool call)" with condition "last_tool_call_time > 30s."

A second transition arrow 208 from the autonomous state 202 to the recovering state 204 is labeled "Cognitive layer reconnects (resync called)" with condition "resync() tool invoked."

A third transition arrow 210 from the recovering state 204 to the connected state 200 is labeled "Resync complete, buffer transmitted" with condition "Cognitive layer acknowledges state."

A fourth transition arrow 212 shown as a dashed line from the autonomous

state 202 directly to the connected state 200 is labeled "Any tool call received (heartbeat reset)" with condition "Direct reconnection without explicit resync."

A self-loop arrow 214 on the autonomous state 202 is labeled "Each autonomous step (up to 10,000)" with condition "autonomous_steps < hard_cap" and annotation "Hard cap: 10,000 steps."

FIG. 3 — Energy-Aware Modulation Curve

FIG. 3 is a graph showing the step function that maps energy level to autonomous modulation value.

An energy modulation curve 300 is plotted with the x-axis labeled "System Energy Level (mWh)" ranging from 0 to 100+ and the y-axis labeled "Autonomous Modulation Value" ranging from -0.55 to +0.45 with tick marks at -0.4, -0.1, 0.0, and +0.3 corresponding to the actual modulation values.

The energy modulation curve 300 comprises four plateaus. A first plateau from 0 to 15 mWh has modulation -0.4, is shaded with dense hatching, and labeled "CRITICAL — Maximum conservation" with behavioral note "Inhibit neural activity to preserve energy." A second plateau from 15 to 30 mWh has modulation -0.1, is shaded with light hatching, and labeled "LOW — Cautious operation" with behavioral note "Slightly reduce activity." A third plateau from 30 to 80 mWh has modulation 0.0, is unshaded, and labeled "NORMAL — Neutral operation" with behavioral note "Standard autonomous processing." A fourth plateau from 80+ mWh has modulation +0.3, is shaded with light hatching, and labeled "SURPLUS — Opportunistic exploration" with behavioral note "Increase activity to utilize excess energy." Vertical dashed lines appear at the energy thresholds of 15, 30, and 80 mWh.

A behavioral effects table 302 appears below the energy modulation curve 300 and annotates: "The system autonomously adjusts its processing intensity

based on available energy — mimicking biological metabolic regulation."

FIG. 4 — Autonomous Input Generator Output

FIG. 4 is a signal diagram showing the time-series waveform of the self-generated input signal.

An input signal graph 400 occupies the upper portion of the figure with

x-axis "Autonomous Step Number" ranging from 0 to 500 and y-axis "Input Value" ranging from 0.0 to 1.0.

The input signal graph 400 contains three superimposed waveform components.

A circadian base signal is shown as a smooth sinusoidal curve with equation

"base = 0.5 + 0.3 x sin(2 x pi x step / period)" with period = 500 steps

and amplitude range 0.2 to 0.8. Superimposed on the base signal are

attention bursts 402 shown as sharp upward spikes occurring randomly at

approximately 10% of steps, each burst 402 adding +0.2 above the base

signal. Each burst 402 is labeled with a small arrow: "10% probability

burst." The composite signal (base plus bursts) is shown as the final bold line labeled "Composite autonomous input."

An autonomous input formula box 404 shows the equation governing the composite signal generation.

Below the input signal graph 400, an energy recovery graph 406 shows the corresponding energy level over the same time period. The energy recovery graph 406 has y-axis "Energy (mWh)" ranging from 0 to 105 and shows a monotonic recovery trajectory starting at 10 mWh in the CRITICAL zone and rising through all four modulation zones to the capacity ceiling of 100 mWh.

Recovery rate in the energy recovery graph 406 is zone-dependent: slow in CRITICAL (0.15 mWh/step), moderate in LOW (0.25 mWh/step), steady in NORMAL (0.40 mWh/step), and slow near SURPLUS (0.10 mWh/step as self-discharge **balances harvest). Three horizontal dashed lines appear at 15, 30, and 80 mWh thresholds with zone labels in the left margin: CRITICAL, LOW, NORMAL,**

SURPLUS.

FIG. 5 — Resynchronization Payload Structure

FIG. 5 is a structural diagram showing the hierarchical data structure of the resync payload.

A resynchronization payload 500 is depicted as a large top-level rectangle containing three main sections.

A summary statistics section 502 is depicted as a rectangle within the resynchronization payload 500. The summary statistics section 502 contains fields: "steps_autonomous: [integer]," "energy_delta: [float] mWh," "min_energy: [float] mWh," "max_energy: [float] mWh," "mean_energy: [float] mWh," "total_spikes: [integer]," and "mean_spikes_per_step: [float]." The summary statistics section 502 is labeled "ALWAYS included (compact summary)" with size annotation "~200 bytes."

An energy delta detail section 504 is depicted as a rectangle within the resynchronization payload 500. The energy delta detail section 504 contains fields: "energy_start: [float] mWh," "energy_end: [float] mWh," and "energy_delta: [float] mWh." An arrow from the energy_delta field leads to annotation: "Positive = system gained energy during autonomy."

A full step buffer section 506 is depicted as a larger rectangle within the resynchronization payload 500 with a repeating row structure showing individual step records: "Step 0: {input, modulation, spikes, energy, temp, pattern}" through "Step N: {input, modulation, spikes, energy, temp, pattern}." The full step buffer section 506 is labeled "OPTIONAL — Full operational history" with size annotation "~100 bytes x N steps" and condition "Included when cognitive layer requests detailed history."

Two granularity levels are annotated: an arrow to the summary statistics

section 502 labeled "Level 1: Summary only (fast resync)" and an arrow to the full step buffer section 506 labeled "Level 2: Full buffer (complete reconstruction)."

A decision diamond 508 at the bottom of the figure asks "Cognitive layer requests full buffer?" A yes arrow leads to "Transmit Level 2 (summary + full buffer)" and a no arrow leads to "Transmit Level 1 (summary only)."

FIG. 6 — Recovery Timeline Diagrams

FIG. 6 shows three horizontal timeline diagrams stacked vertically, each illustrating a different recovery scenario.

A normal reconnection timeline 600 depicts the first scenario. The normal reconnection timeline 600 progresses from left to right through five phases: Connected (Claude active), Timeout (watchdog counting for 30 seconds), Autonomous (self-generated input plus energy-aware modulation), Resync (buffer transmitted), and Connected again (cognitive control restored).

Labels on the normal reconnection timeline 600 include: "Claude goes silent" at the transition to timeout, "Fallback engages" at the transition to autonomous mode, "resync() called" at the transition to resync, and "Seamless resumption" at the return to connected mode. A buffer icon indicates "N steps buffered."

A crash recovery timeline 602 depicts the second scenario. The crash recovery timeline 602 progresses through: Connected (normal operation), CRASH (MCP server terminates), Server Restart (process relaunched), Snapshot Load (reads latest.json), and Connected (state restored). Labels on the crash

recovery timeline 602 include: "Server process dies" at the crash point, "Process relaunched" at restart, "initialize_consciousness loads snapshot" at snapshot load, and annotation "State restored from last snapshot (max 50 steps lost)." A snapshot icon indicates "latest.json (serialized every 50 steps)."

A clean shutdown timeline 604 depicts the third scenario. The clean shutdown timeline 604 progresses through: Connected (normal operation), Autonomous (fallback operation), EOF Signal (stdin closed), and Shutdown (final snapshot **written to disk). Labels on the clean shutdown timeline 604 include: "Claude disconnects," "Process receives EOF," "finally block writes snapshot," and annotation "State preserved for next session."**

A recovery guarantees box 606 provides common annotations for all three timelines: solid line segments indicate "System actively processing," dashed line segments indicate "System in transition," and bold markers at each state change indicate "No data loss at any transition."

FIG. 7 — End-to-End Signal Flow: Connected vs. Autonomous Operation

FIG. 7 is an end-to-end signal flow diagram showing the complete system in a split-view format, with connected operation on top and autonomous operation on bottom, sharing the same core processing pipeline.

In the top half labeled "CONNECTED MODE," a cognitive layer 700 is depicted as an ellipse or cloud shape at top. Dashed arrows from the cognitive layer 700 carry cognitive modulation commands downward to the processing pipeline.

The connected-mode processing pipeline flows left to right: sensors 702 receive environmental input, feed into SNN processing 704, which feeds into cognitive modulation 706 (labeled "External AI sets reflection_coeff, modulation"), which feeds into motor actuation 708. An energy feedback loop 710 returns from the motor actuation 708 back to the SNN processing 704 as bold dashed arrows, labeled "Patent A." A self-observation feedback path (dotted arrows) feeds SNN output back to the SNN processing 704. The label "Heartbeat active (tool calls within 30s)" indicates normal connected operation.

A horizontal dashed dividing line across the full width is labeled "DISCONNECTION EVENT (heartbeat timeout > 30s)" with an arrow pointing down

labeled "Seamless transition — no interruption to SNN or Harvester."

In the bottom half labeled "AUTONOMOUS MODE," an autonomous input generator

712 replaces the sensors 702, containing "Circadian base signal" and "10% stochastic attention bursts." The autonomous-mode processing pipeline flows: autonomous input generator 712 feeds into SNN processing 704, which feeds into energy-aware self-modulation 714 (labeled "Internal energy-aware regulation" with four modulation regions: "<15 mWh: -0.4, <30 mWh: -0.1, 30-80 mWh: 0.0, >80 mWh: +0.3"), which feeds into motor actuation 708. The energy feedback loop 710 returns from motor actuation 708 back to SNN processing 704 as bold dashed arrows. A self-observation feedback path (dotted arrows) feeds SNN output back to SNN processing 704.

Additional components in autonomous mode include a state buffer 716 depicted as a cylinder or database shape, receiving every step from SNN processing

704 with label "Rolling buffer (max 10,000 entries)." A snapshot writer 718 depicted as a disk shape receives data from the state buffer 716 every 50 steps with label "latest.json (persistent storage)."

On the right side of the figure, a resync path 720 shows an arrow from the bottom half back to the top half, labeled "resync() called." The resync path 720 includes a rectangle labeled "Resync Payload" containing "steps_autonomous, energy_delta, summary" and "Optional: full step-by-step buffer," with an arrow upward to the cognitive layer 700 labeled "Cognitive layer catches up."

The SNN processing 704, motor actuation 708, and energy feedback loop 710 are shared components drawn once spanning both halves, with label "Core pipeline never stops."

Key annotations state: "The fallback system changes the INPUT SOURCE and

MODULATION SOURCE. The core neural processing, motor actuation, and energy harvesting continue WITHOUT INTERRUPTION through the transition."

A legend box distinguishes: solid arrows for forward signal path, bold dashed for energy feedback loop, dotted for self-observation feedback loop, and thin dashed for cognitive/modulation control path. "The transition between modes changes only the control paths, never the core processing paths."

Sheet numbering: "7/7" at top center

GENERAL DRAWING REQUIREMENTS (per 37 CFR 1.84)

- Paper size: 8.5 x 11 inches (US Letter)

- Top margin: 1 inch

- Left margin: 1 inch

- Right margin: 5/8 inch

- Bottom margin: 3/8 inch

- Black lines on white background ONLY

- No color, no grayscale shading (use hatching patterns instead)

- Line weight: sufficiently heavy for adequate reproduction

- Each figure labeled: "FIG. 1", "FIG. 2", etc. - Sheet numbers at top center: "1/7", "2/7", etc.

- No frames or borders around drawing area

- All text in figures must be at least 1/8 inch (3.2 mm) high

- Reference numerals must be used consistently across all figures